Cyrus Yong

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Education

Georgia Institute of Technology Atlanta, GA

B.S. Computer Science & B.S. Electrical Engineering | Minor: Robotics | GPA: 3.6/4.0 *Aug 2024 – May 2028*

- EE Concentrations: Robotics & Autonomous Systems, Circuit Technology | Dean's List: Fall 2024, Spring 2025

Georgia State University Atlanta, GA

B.S. Computer Science (Transferred to Georgia Tech) | GPA: 3.7/4.0 *Aug 2022 – May 2024*

- President's List: Spring 2023, Spring 2024 | Dean's List: Fall 2023

Experience

Georgia Tech Research Institute (GTRI) Atlanta, GA

Software Engineering Intern, ASL *Aug 2026 – Present*

- Set up a paired Windows and Linux (RHEL) VM environment for a C++ simulation project: RDP'ing into both machines, cloning and building the codebase on Linux, and configuring networking between the VMs.
- Rewriting C++ message handlers for the TSS communication protocol, pulling messages off a deque, validating type, and routing to the application manager, while eliminating duplicate `memcpy` logic between paired handlers and reverse-engineering legacy code to query an external device for outbound message bodies.

Intelligent Platform for Crowdsourcing (IPC) — VIP — Github Atlanta, GA

Undergraduate Researcher, Georgia Tech *Aug 2025 – May 2026*

- Built a structured data-extraction pipeline using LiteLLM on GT's PACE GPU cluster, parsing unstructured input into defined fields for downstream workflows.
- Developed a RAG-powered chatbot backend with controlled retrieval from curated resources, reducing hallucinations and improving system reliability.
- Architected a React + FastAPI full-stack application with custom document ingestion (BeautifulSoup, PyPDF) for knowledge-base construction.

Projects

STM32 Vending Machine — Embedded Control System *June 2026 – Present*

C, STM32 Nucleo-F446RE (ARM Cortex-M4), FSM, PWM Servos, GPIO EXTI, I^2C, SPI/FatFS

- Building C firmware on an STM32 Nucleo-F446RE (ARM Cortex-M4) driven by a finite state machine that handles push-button navigation (GPIO EXTI), payment logic, and PWM control of MG995 servo gates across three drink stacks.
- Extending the design with IR break-beam sensors for coin detection and drop confirmation, an SSD1306 OLED menu (I^2C), and DS3231 RTC-timestamped transaction logging to microSD over SPI (FatFS).

STM32 Temperature Data Logger — Embedded Firmware — Github *June 2026 – Present*

C, STM32 Nucleo-F446RE (ARM Cortex-M4), BME280 (I^2C), microSD (SPI), FatFS

- Developing C firmware on an STM32 Nucleo-F446RE (ARM Cortex-M4) to sample a BME280 sensor (temperature, humidity, pressure) over I^2C at configurable intervals.
- Logging timestamped readings to a FAT32 microSD card over SPI (FatFS) for offline retrieval and analysis, implementing peripheral drivers and buffering on bare metal.

USB HID Macropad — Custom Firmware & PCB — Github *May 2026 – Present*

C (bare-metal), Raspberry Pi Pico (RP2040), TinyUSB, KiCad, 3D Printing

- Developing bare-metal C firmware on the RP2040 that enumerates as a USB HID composite device, handling descriptor setup and key-event reporting through the TinyUSB stack.
- Programmed each key to emit a predefined multi-keystroke macro over USB HID, with a custom KiCad PCB and 3D-printed enclosure, building on a prior single-button HID controller.

LTC2308 ADC Peripheral for SCOMP — ECE 2031 Final Project *Jan 2026 – May 2026*

VHDL, SPI, DE10-Standard FPGA, Quartus Prime, SCOMP Assembly

- Designed a custom VHDL I/O peripheral enabling a 16-bit SCOMP processor to read analog data from an 8-channel, 12-bit LTC2308 ADC over SPI.
- Wrote a 5-state FSM managing SPI timing at 25 MHz, shifting 6-bit channel-config words on SDI while capturing 12-bit results on SDO, behind an 8-register memory-mapped I/O interface.

Technical Skills

Embedded & Hardware: C (bare-metal), C++, Embedded Firmware, STM32 (ARM Cortex-M), Raspberry Pi Pico (RP2040), TinyUSB / USB HID, VHDL, FPGA (Quartus), SPI, I^2C, UART/Serial, PWM, GPIO/Timers, SD Card / FatFS, Arduino, KiCad (PCB Design), Fusion 360, 3D Printing, Oscilloscope, Logic Analyzer

Languages: Python, C, C++, Java, JavaScript, TypeScript, VHDL, Assembly (SCOMP), HTML/CSS

Software & Tools: FastAPI, ReactJS, LiteLLM, OpenCV, Git, Linux, AWS (Lambda, S3, API Gateway), Firebase

Relevant Coursework: Digital Design Lab (ECE 2031), Digital Logic Design (ECE 1100), Circuit Analysis, Computer Organization & Programming, Systems & Networks, Computer Vision (CS 4476), Machine Learning